Eduqas GCSE Geography (C/4) — Component 1

# 1.2 River Landscapes

## 1. The River as a System

A river and its drainage basin function as an **open system** — water, sediment, and energy enter, are processed, and leave. Understanding this systems framework underpins all river processes and landforms.

### Key Drainage Basin Terms

**Drainage basin:** The area of land drained by a river and its tributaries. Also called a **catchment area**. It is bounded by a **watershed** (the ridge of high land separating one drainage basin from another).

**Watershed:** The boundary (usually a ridge) separating one drainage basin from another. Precipitation falling on one side flows into a different river system.

**Tributary:** A smaller stream or river that joins a larger one.

**Confluence:** The point where two rivers or streams join.

**Source:** The origin (usually highest point) of a river, often a spring, bog, or lake.

**Mouth:** Where the river meets the sea or a lake. Discharge is greatest here.

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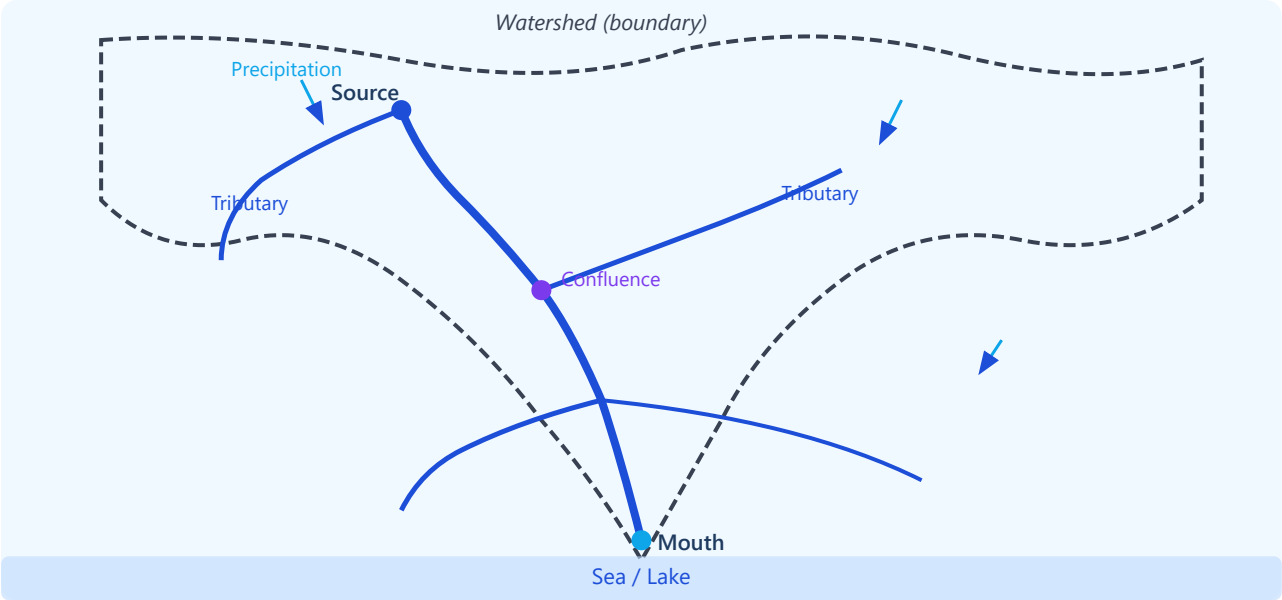


*Fig 2.1 — A drainage basin showing watershed, source, tributaries, confluence, and mouth*

## Inputs, Throughputs, and Outputs

| Component | Description |
|---|---|
| **Input** | Precipitation (rain, snow, sleet) — enters the drainage basin as water |
| **Infiltration** | Water soaking into the soil; rate depends on soil type, saturation, vegetation |
| **Surface runoff (overland flow)** | Water flowing over land surface when infiltration capacity is exceeded; fast — reaches river quickly |
| **Throughflow** | Water moving through the soil layer, parallel to the slope; moderate speed |
| **Groundwater flow** | Slow movement of water through permeable rock; sustains rivers during dry spells (base flow) |
| **Output: River discharge** | Volume of water passing a point per second (cumecs, $m^3/s$); measured at gauging stations |
| **Output: Evapotranspiration** | Water lost to atmosphere through evaporation + plant transpiration; higher in summer |

# 2. River Processes

## Erosion

**Hydraulic Action:** The force of flowing water pushing into cracks and joints in the river bed and banks. The pressure compresses air, weakens rock, and can dislodge large blocks. Most effective during floods.

**Abrasion (Corrasion):** The load carried by the river (pebbles, sand) acts like sandpaper, scraping and grinding the bed and banks. Responsible for vertical and lateral erosion; creates smooth, polished channels.

**Attrition:** Particles of load collide with each other, becoming progressively smaller, rounder, and smoother as they travel downstream. Explains why large angular boulders near the source become small, round pebbles near the mouth.

**Corrosion (Solution):** Slightly acidic river water chemically dissolves soluble minerals (e.g., calcium carbonate in limestone). The dissolved material is carried in solution. Example: limestone karst landscapes.

Rivers also erode in different directions: **vertical erosion** (downward, deepening the channel — dominant in upper course) and **lateral erosion** (sideways, widening the channel — dominant in middle and lower course).

## Transportation

| Process | Sediment type | Description |
|---|---|---|
| **Traction** | Boulders, cobbles | Largest material rolled along the river bed by the current |
| **Saltation** | Pebbles, coarse sand | Bounced along the bed in a series of hops |

| | | |
|---|---|---|
| **Suspension** | Fine silt, clay | Carried within the water column; gives rivers brown colour after heavy rain |
| **Solution** | Dissolved minerals | Carried invisibly in the water; always in transport regardless of velocity |

### Deposition

Deposition occurs when a river loses **energy** (velocity decreases) and can no longer carry its load. This happens when: gradient decreases (entering a flood plain), discharge falls (drought, abstraction), the river meets the sea (loses velocity), or on the inside of meander bends (reduced velocity). The largest/heaviest material is deposited first.

### The Hjulström Curve

The Hjulström curve shows the relationship between water velocity and sediment behaviour (erosion, transport, deposition):

- **Erosion velocity** — the minimum velocity needed to pick up (entrain) sediment of a given size. Coarser sediment requires more velocity; but importantly, fine clays also need relatively high velocity because they are cohesive (stuck together).
- **Settling/deposition velocity** — the velocity at which a particle is deposited. This is always lower than erosion velocity.
- **Key point:** Once in transport, fine silt/clay can be carried at very low velocities. Large boulders require very high velocities to be entrained, but once moving, they are deposited as soon as velocity drops below transport velocity.

**Exam Tip:** The Hjulström curve is tricky — remember that fine clays require a surprisingly high velocity to erode (because they are cohesive), but once in suspension they stay in transport at very low velocities. For boulders: high velocity to move, and deposited rapidly when velocity drops.

## 3. The Long Profile and Changes Downstream

The **long profile** of a river is a cross-section from source to mouth, showing how gradient changes. Near the source, the gradient is steep; by the lower course, it has almost flattened out, forming a smooth

concave curve called the **graded profile**.

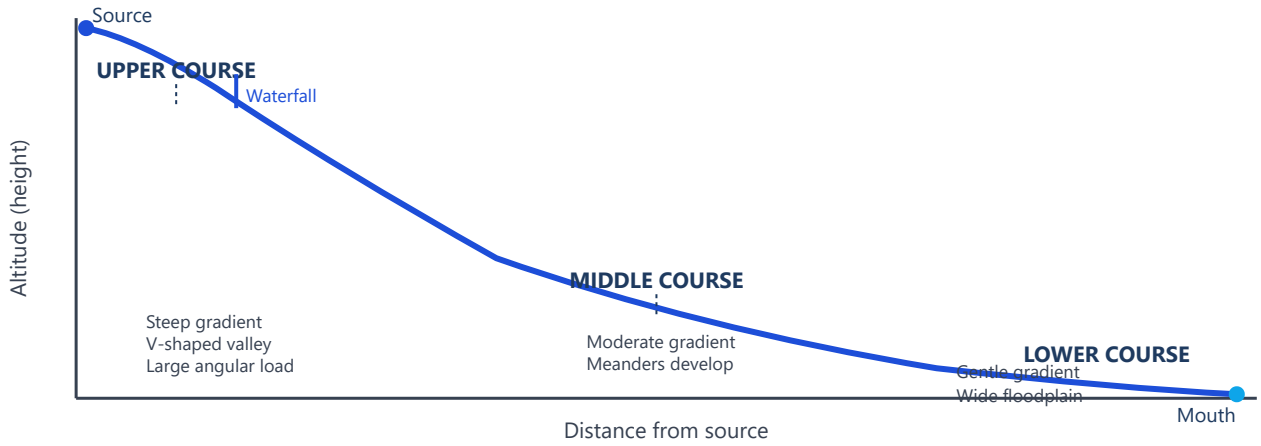


*Fig 2.2 — River long profile (concave shape) showing upper, middle, and lower course characteristics*

| Variable | Upper Course | Middle Course | Lower Course |
|---|---|---|---|
| Gradient | Steep | Moderate | Gentle/flat |
| Velocity | Low (high friction) | Increasing | Fastest overall |
| Discharge | Lowest | Increasing (tributaries) | Highest |
| Channel width | Narrow | Wider | Widest |
| Channel depth | Shallow | Deeper | Deepest |
| Load size | Large, angular boulders | Pebbles, gravels | Fine sand, silt |
| Dominant erosion | Vertical (deepening) | Lateral (widening) | Lateral (very wide) |
| Valley shape | V-shaped, narrow | Wider, flatter floor | Broad, flat floodplain |

# 4. Upper Course Landforms

## V-Shaped Valleys and Interlocking Spurs

In the upper course, the river mainly erodes **vertically**, cutting downward into the bedrock. The valley sides are steep because:

- Rainwater washes material off steep valley sides (mass movement and surface wash)
- The valley floor is narrow — little lateral erosion
- Weathering (freeze-thaw) loosens material on the valley sides which falls into the river

The resulting valley has a classic **V-shape**. The river winds around resistant spurs of rock, which jut into the valley from alternate sides — these are called **interlocking spurs**.

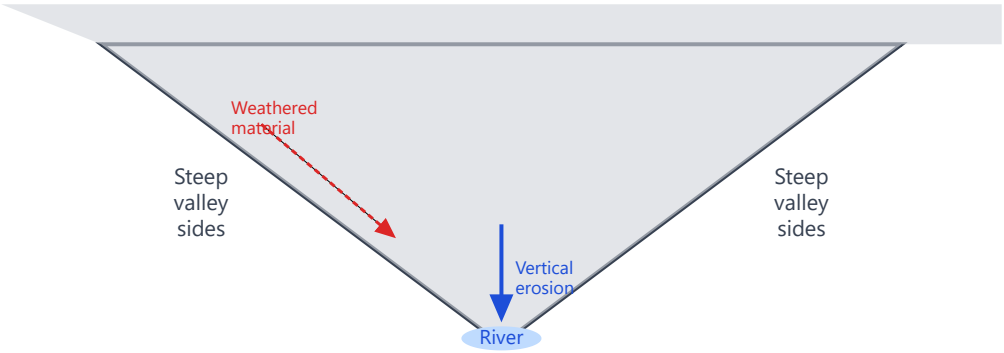


*Fig 2.3 — V-shaped valley cross-section: vertical erosion by river + weathering of valley sides*

## Waterfalls and Gorges

Waterfalls form where a river crosses a band of **resistant rock** overlying softer rock. The sequence of formation:

1. The river flows over the hard rock cap; beneath, softer rock erodes faster by hydraulic action and abrasion
2. A **plunge pool** forms at the base where the falling water churns the rock
3. The hard cap rock is undercut and becomes unsupported; eventually it collapses
4. The waterfall retreats upstream, leaving a steep-sided **gorge** in its wake

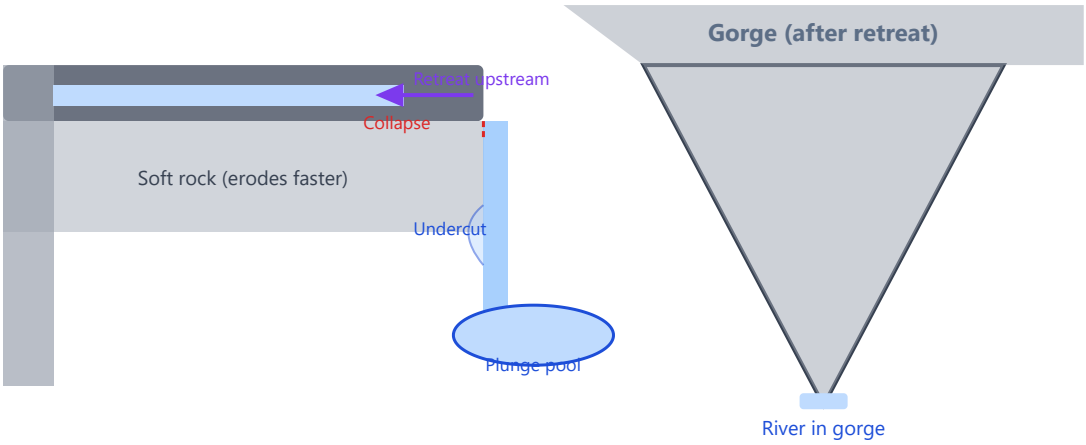


*Fig 2.4 — Waterfall formation and retreat, leaving a gorge. Example: High Force, River Tees, County Durham*

**Example:** High Force, River Tees — the UK's largest waterfall by volume. Hard Whin Sill dolerite overlies soft Carboniferous limestone; the waterfall has retreated ~100 m to form a short gorge.

# 5. Middle Course Landforms

## Meanders

As the river enters the middle course, the valley widens and the river begins to swing in wide bends called **meanders**. They develop because of differences in velocity across the channel:

- On the **outer bend** — faster flow (greater centrifugal force), erosion by abrasion and hydraulic action creates a steep **river cliff** (cut bank)
- On the **inner bend** — slower flow, deposition builds up a **slip-off slope** (point bar) of sand and gravel
- A secondary **helical (corkscrew) flow** called **helicoidal flow** moves water and sediment from the outer bank to the inner bend, enhancing the asymmetric shape

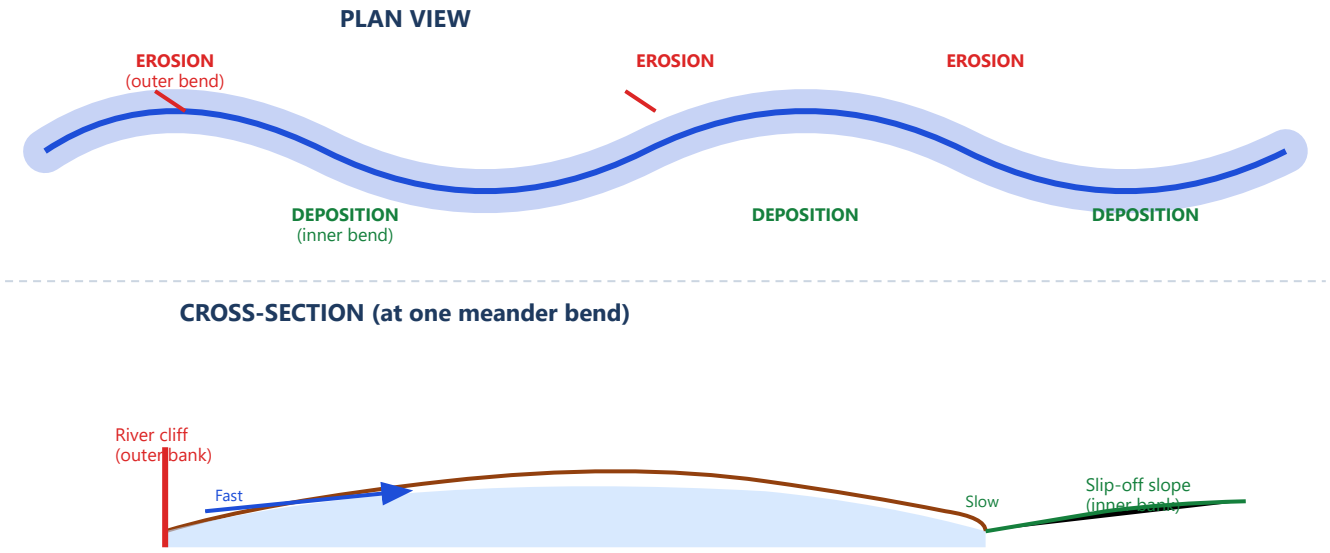


*Fig 2.5 — Meander plan view (top) and cross-section (bottom) showing river cliff on outer bank and slip-off slope on inner bank*

# 6. Lower Course Landforms

## Ox-bow Lakes

Ox-bow lakes form from meanders through a 4-stage process:

1. The meander bends become increasingly tight (tortuous) as erosion continues on outer bends
2. During a flood, the river cuts through the narrow neck of land between two adjacent meander bends (the shortest route)
3. The river straightens, depositing sediment (alluvium) at both ends of the old meander loop
4. The abandoned meander bend is cut off, forming a horseshoe-shaped **ox-bow lake**
5. Over time, the ox-bow dries up, leaving a curved **meander scar**

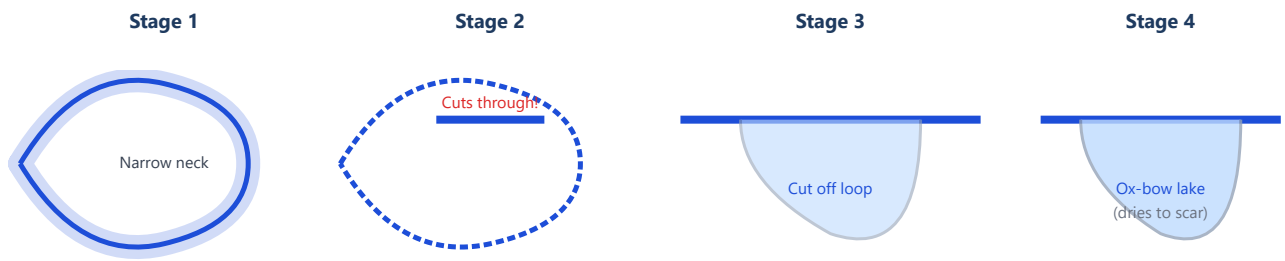


*Fig 2.6 — Four-stage formation of an ox-bow lake from a meander*

## Flood Plains

A **flood plain** is the flat, low-lying area either side of the river channel, formed by repeated flooding and lateral erosion. During floods, the river overtops its banks and velocity drops immediately — sediment (alluvium) is deposited across the flood plain, gradually building it up. Flood plains are the most fertile agricultural land due to mineral-rich alluvium.

### Levées

Natural embankments alongside river channels. During floods, coarsest sediment is deposited immediately adjacent to the channel (as velocity falls sharply at the bank). Over many flood events, these deposits build up to form **levées**. They may raise the channel above the level of the surrounding flood plain. When levées breach (break), flooding can be catastrophic (e.g., Yellow River, China; Mississippi, USA).

### Deltas

Deltas form at river mouths where sediment load is deposited faster than it can be removed by tidal currents or waves. The river splits into distributaries. Two main types:

- **Arcuate delta** — fan-shaped, smoothed by wave action. Example: Nile Delta, Egypt
- **Bird's foot delta** — long, finger-like distributaries extending into calm water. Example: Mississippi Delta, USA

## 7. Storm Hydrographs

A **storm hydrograph** shows how river discharge responds to a rainfall event over time. It is a key tool for flood prediction and management.

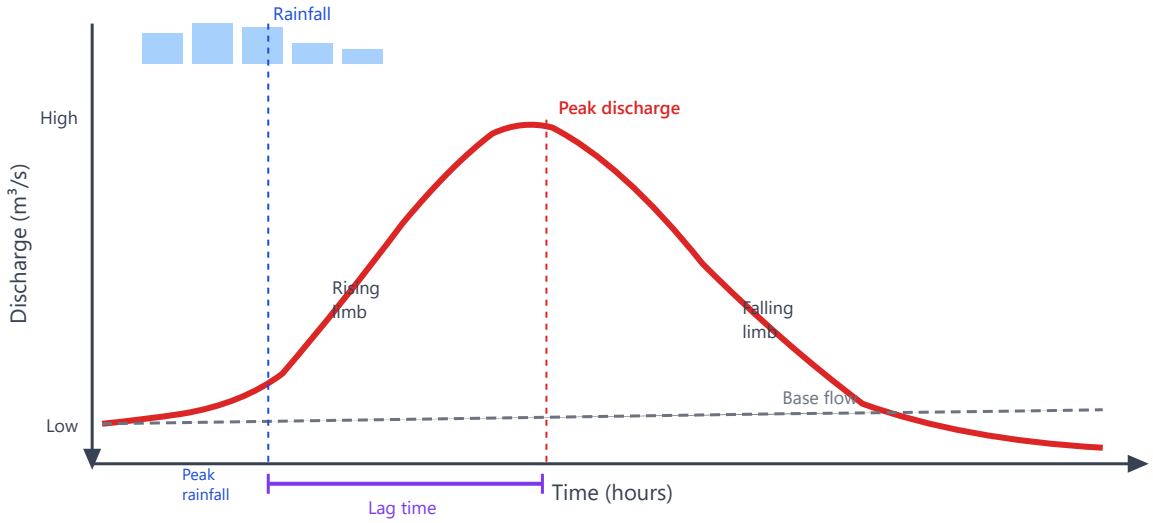

*Fig 2.7 — A storm hydrograph showing key components*

## Key Components

| Term | Definition |
|---|---|
| **Rising limb** | The rising section of the graph as discharge increases after rainfall |
| **Peak discharge** | The maximum discharge reached after a storm event ($m^3/s$) |
| **Lag time** | The time delay between peak rainfall and peak discharge; shorter = flashier = more flood risk |
| **Falling limb (recession limb)** | The gradual decrease in discharge as water drains away |
| **Base flow** | The background level of discharge sustained by groundwater between storms |

## Factors Affecting Hydrograph Shape

| Factor | Flashy hydrograph (short lag, high peak) | Subdued hydrograph (long lag, low peak) |
|---|---|---|
| Land use | Urban areas (impermeable tarmac, drains) | Forest/pasture (infiltration, interception) |
| Rock type | Impermeable rock (granite, clay) | Permeable rock (chalk, sandstone) |
| Rainfall intensity | Heavy, intense, short storm | Light, prolonged drizzle |
| Vegetation | Bare soil, arable fields, deforested | Dense woodland, grassland |
| Slope angle | Steep slopes — rapid runoff | Gentle slopes — slow runoff |
| Antecedent soil moisture | Already saturated soil — rapid runoff | Dry soil — high infiltration capacity |

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**Exam Tip:** When describing a hydrograph, always quote figures: "peak discharge of X $m^3/s$", "lag time of Y hours". When explaining hydrograph shape, give reasons linked to the specific drainage basin characteristics — e.g., "the short lag time of 2 hours suggests an urbanised catchment with impermeable surfaces and efficient drainage systems."

# 8. Causes and Effects of Flooding

## Physical Causes

- **Prolonged rainfall** — saturates soil, reducing infiltration; surface runoff increases rapidly
- **Heavy, intense storms** — rainfall exceeds infiltration capacity immediately
- **Snowmelt** — rapid melting in spring (especially when combined with rain) releases large volumes of water
- **Impermeable rock** (granite, clay) — little water infiltrates, most runs off
- **Steep slopes** — water reaches the river channel quickly
- **Low vegetation cover** — less interception and transpiration

## Human Causes

- **Urbanisation** — tarmac, concrete, and rooftops are impermeable; drains deliver water rapidly to rivers, reducing lag time and increasing peak discharge
- **Deforestation** — trees intercept rainfall and roots absorb water; removing trees increases runoff and reduces evapotranspiration
- **Floodplain development** — building on flood plains removes natural storage capacity and increases flood risk downstream
- **Overgrazing and compaction** — compacted soils reduce infiltration rates
- **Climate change** — more intense rainfall events; changing precipitation patterns

**Case Study: Boscastle Flash Flood, Cornwall, August 2004**

**Event:** On 16 August 2004, exceptionally heavy convective rainfall (200 mm in ~3 hours — a 1-in-400-year event) fell over the small drainage basin of the River Valency and River Jordan above Boscastle village.

**Physical causes:**

- Intense, localised convectional rainfall triggered by a stationary low-pressure system on a hot August day
- Steep valley sides and impermeable Devonian slate bedrock — rapid surface runoff
- Small, steep drainage basin with short lag time
- Soils already saturated from previous days of rain

**Effects:**

- Flood peak estimated at 140 $m^3/s$ (normally ~1 $m^3/s$) — a 140-fold increase
- Floodwaters reached 3 m deep in the narrow village street
- Over 100 cars swept out to sea; 6 footbridges destroyed; B3263 road badly damaged
- 58 properties destroyed or severely damaged; 4 businesses lost
- No deaths (thanks to emergency services), but 1 air ambulance rescue; over 100 people rescued by helicopter

**Management responses:**

- The valley was redeveloped with a flood alleviation scheme: river channel widened and deepened; culverts removed; car park relocated to higher ground
- New footbridge designed to allow debris to pass through
- Costs: ~£10 million for reconstruction + £1.5 million for the flood scheme

# 9. River Management Strategies

## Hard Engineering

| Strategy | Description | Advantages | Disadvantages |
|---|---|---|---|
| **Dams and reservoirs** | Large concrete structures built across river valleys; reservoir | Highly effective flood control; HEP generation; water supply; recreation | Extremely expensive; displaces communities; traps sediment (delta starvation downstream); |

| | | | |
|---|---|---|---|
| | stores water, controlling release | | dam failure catastrophic; destroys habitats |
| **Channel straightening (channelisation)** | Meanders cut off to create a straight channel; may line channel with concrete | Water removed faster; reduces local flooding | Increases flooding downstream (water arrives faster); destroys habitat; unnatural; channel can re-meander |
| **Embankments (artificial levées)** | Raised earthen or concrete banks along river channel to contain flood water | Relatively cheap; can raise height over time; allows floodplain use | If breached, flooding is catastrophic; increases flood risk downstream; regular maintenance needed |
| **Flood relief channels** | Parallel channels dug to carry overflow when main channel is in flood | Effective; can be designed to specific discharge; reduces pressure on main channel | Land take; expensive to construct; only effective if correctly sized for design flood |

## Soft Engineering

| Strategy | Description | Advantages | Disadvantages |
|---|---|---|---|
| **Afforestation** | Planting trees in the drainage basin to increase interception and infiltration | Sustainable; cheap; improves biodiversity; carbon sequestration; reduces runoff long-term | Takes decades to be effective; may reduce agricultural productivity of upland areas |
| **Floodplain zoning** | Planning regulations prevent development on flood-risk areas; land left as parks, recreation | Prevents building in flood zones; relatively cheap; maintains natural floodplain function | Restricts development in areas where land is scarce; planning enforcement required; does not reduce existing flood risk |
| **Washlands / water meadows** | Designated areas of floodplain deliberately allowed to flood to | Free; natural; reduces flood peak; maintains | Takes agricultural land out of production; only useful for |

| | | | |
|---|---|---|---|
| | temporarily store excess water | biodiversity; no maintenance needed | moderate floods; landowner compensation needed |

**Exam Tip:** For "evaluate" questions on river management, always consider: **effectiveness, cost, sustainability, impact on people and environment, and downstream effects**. Hard engineering tends to be effective short-term but can transfer the problem downstream. Soft engineering is more sustainable but may take time to work and requires large areas of land.

## 10. Case Study — Bangladesh Flooding

**Case Study: Bangladesh — Causes, Effects, and Management of Flooding**

**Background:** Bangladesh is one of the world's most flood-prone countries. Around **80% of the country is a flood plain** (the combined Ganges-Brahmaputra-Meghna delta). Each year, 20–30% of the country floods during the monsoon season (June–September); in severe years (e.g., 1998, 2017), 60–70% is submerged.

**Physical causes:**

- Three major rivers (Ganges, Brahmaputra, Meghna) converge in Bangladesh, bringing meltwater from Himalayan glaciers and monsoon rain from across South Asia
- **Snowmelt** in spring adds to river discharge before the monsoon even begins
- **Monsoon rainfall** — Bangladesh receives 1,200–2,500 mm/year, mostly June–September; very intense daily totals possible
- Flat, low-lying land — much of Bangladesh is <1 m above sea level; water cannot drain away
- Cyclones (e.g., Cyclone Sidr 2007) bring storm surges and coastal flooding from the Bay of Bengal

**Human causes:**

- **Deforestation** in Nepal and the Himalayas has increased surface runoff and sediment load into rivers
- Population growth and urbanisation on floodplain land increases flood vulnerability
- **Climate change** — rising sea levels (Bangladesh is predicted to lose 17% of land area by 2050 with a 1 m rise) and more intense storms
- Upstream water management (dams in India) can cause unpredictable water releases

**Effects of flooding (e.g., 1998 floods — the most severe of the 20th century):**

- 66% of Bangladesh submerged for 2–3 months
- 30 million people made homeless
- 1,100 deaths from drowning, disease (cholera, typhoid), and snake bites
- 700,000 ha of cropland destroyed — food security crisis; rice prices rose by 30%
- ~$1 billion damage to infrastructure (roads, bridges, schools, clinics)
- Dhaka (capital city) partly submerged; factories closed, economic losses severe

**Positive effects:**

- Annual flooding deposits rich alluvium, making Bangladesh's soils extremely fertile — foundation of its rice agriculture
- Recharges groundwater aquifers

**Management responses:**

- **Flood Action Plan (FAP)** — funded by World Bank; involves building 7,000 km of embankments along rivers. Controversial: embankments can fail catastrophically; traps water inside; disrupts fishing; does not address causes
- **Flood Forecasting and Warning Centre (FFWC)** — river gauge monitoring and satellite data used to give 72-hour flood forecasts; allows evacuation of vulnerable areas
- **Flood shelters** — raised concrete cyclone shelters provide refuge for 1–2 million people during flooding; particularly important in coastal zones
- **BRAC (NGO)** — community-level flood preparedness: waterproof storage of food, seeds, and documents; floating gardens (baira) for food production during floods
- **Flood-resistant housing** — houses built on raised platforms (plinth) using earth mounds; bamboo and concrete designs that withstand moderate flooding

**Evaluation:** Hard engineering (embankments) has had mixed results — protecting some areas while increasing risk in others. Community-based approaches (early warning, shelters, flood-resistant crops) are increasingly recognised as more sustainable for a country that cannot afford to prevent flooding and must learn to live with it. Climate change and sea-level rise make long-term management increasingly difficult.

**Exam Tip:** Bangladesh is a superb multi-mark case study because it links physical geography (rivers, monsoon, delta), human geography (poverty, population), and global issues (climate change, sea-level rise, deforestation). Always include statistics: "66% of Bangladesh flooded in 1998", "30 million homeless", "80% of land is floodplain". Generic answers without data score in band 1 only.